

Soft-bodied animals

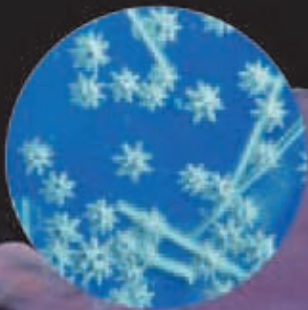
Around 700 million years ago, the world's first animals lived in water and had no hard body parts. Today, soft-bodied animals still thrive in the oceans and in many other wet or damp habitats, from lakes and rivers to soil. They have different ways of making their soft bodies rigid, which gives their muscles something to pull against when they move. Many soft-bodied animals are microscopic, but the largest include jellyfish more than 6 ½ ft (2 m) across.

Swimming bell

A jellyfish's umbrella-shaped body, or bell, is made of a solid jellylike substance covered by a thin layer of living cells. The mouth and stomach cavity are in the center of the bell, surrounded by fleshy arms.

The jellyfish swims by tensing and relaxing a ring of muscle cells around its central opening. It draws water in, then forces it out of its bell, propelling itself along.

Needle- and star-shaped sponge spicules



Hidden strength

Sponges are very simple animals that feel soft to the touch, but if you squeeze one, it soon springs back into shape. Sponges do this because they are reinforced with microscopic mineral crystals, or spicules, set in an elastic substance called spongin. This elephant ear sponge has spicules made of silica, the same mineral used to make glass. Sponges live by pumping water through holes known as pores. As the seawater flows through their pores, they filter out any food that it contains.

Stinging tentacles catch prey